

Active set method for convex quadratic problem (0.75 pt)

Consider the following 2-dimensional minimization problem

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 \\ \text{subject to:} \quad & x-2y+2 \geq 0, \quad (C_1) \\ & -x-2y+6 \geq 0, \quad (C_2) \\ & -x+2y+2 \geq 0, \quad (C_3) \\ & x \geq 0, \quad (C_4) \\ & y \geq 0 \quad (C_5) \end{aligned}$$

use the active set method for convex QPs starting from the following points:

1. $(x^0, y^0) = (1, 1/2)^T$ for $\mathcal{A}^0 = \emptyset$
2. $(x^0, y^0) = (3, 1/2)^T$ for $\mathcal{A}^0 = \{3\}$
3. $(x^0, y^0) = (0, 1/2)^T$ for $\mathcal{A}^0 = \{4\}$

Sketch the feasible set and for every starting point:

1. Sketch the path of the active set method.
2. Verify the path of the active set method computationally.
Note for each step the following states in the table below:

- the primal variables before the step (x_k, y_k)
- the active set before the step A_k
- the primal variables after the step (x_{k+1}, y_{k+1})
- the dual variables after the step λ_{k+1} , e.g. $(0, 1, -1, 0, 0)$
- the active set after the step A_{k+1}
- whether the KKT conditions are satisfied for $(x_{k+1}, y_{k+1}, \lambda_{k+1})$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	$(1, 1/2)$	$\emptyset$				
1						
2						
3						

[Hint: See Nocedal example 16.4 p. 475. and exercise 16.10]

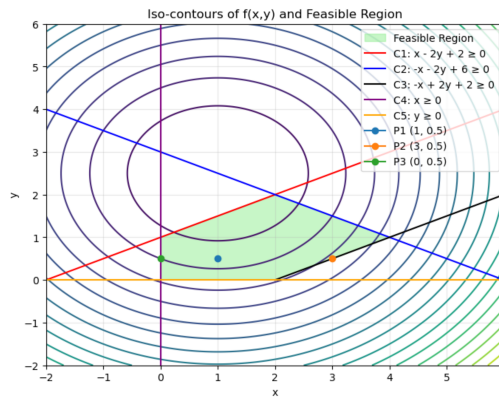
First, I will rewrite the optimization problem with the inequality constraints in standard form $(c_i \leq 0)$.

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 \\ \text{subject to} \quad & -x+2y-2 \leq 0 \quad (c_1), \\ & x+2y-6 \leq 0 \quad (c_2), \\ & x-2y-2 \leq 0 \quad (c_3), \\ & -x \leq 0 \quad (c_4), \\ & -y \leq 0 \quad (c_5) \end{aligned}$$

I will also write each constraint in the form $a_j^T x \leq b_j$:

$$\begin{aligned} a_1 &= \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b_1 = 2 & a_4 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad b_4 = 0 \\ a_2 &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad b_2 = 6 & a_5 &= \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad b_5 = 0 \\ a_3 &= \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad b_3 = 2 \end{aligned}$$

Here we have the visualization of the optimization problem:



$f(x) = (x-1)^2 + (y-2.5)^2$ can be written in QP form $\frac{1}{2}z^T Q z + q^T z + \text{const}$

using: $Q = 2I_2$ and $q = \begin{bmatrix} -2 \\ -5 \end{bmatrix}$.

Also $\nabla f(x) = [2(x-1), 2(y-2.5)]$.

Let's apply Active-set method for each case:

1. $(x^0, y^0) = (1, 1/2)^T$ for $A^0 = \emptyset$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

As $A^0 = \emptyset$ B_0 and b_0 are 0:

$$Q \cdot x_1 = -q \Rightarrow x_1 = -Q^{-1}q = -\frac{1}{2}q = -\frac{1}{2} \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2.5 \end{bmatrix}$$

2) Check for inactive blocking constraints.

C1 and C2 are violated because $a_1^T x_{min} > b_1$ and $a_2^T x_{min} > b_2$.

Step lengths: $\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)}$

$$\alpha_1 = \frac{2 - [-1 \ 2] \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}{[-1 \ 2] \left(\begin{bmatrix} 1 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \right)} = \frac{2 - 0}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{2}{4} = 0.5$$

$$\alpha_2 = \frac{6 - [-1 \ 2] \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{6 - 2}{4} = 1$$

C1 is the first blocking constraint.

3) Update A_1 and x_1 :

$$x_1 = x_0 + \alpha_1 (x_1 - x_0) = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 0.5 \left(\begin{bmatrix} 1 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A_1 = A_0 \cup \{1\} = \{1\}$$

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

$$\begin{cases} Qx_2 + \begin{bmatrix} -1 \\ 2 \end{bmatrix} \lambda = -q \\ [-1 \ 2] x_2 = b_1 \end{cases} \Rightarrow \begin{cases} 2x_2^1 - \lambda_2 = 2 \\ 2x_2^2 + 2\lambda_2 = 5 \\ -x_2^1 + 2x_2^2 = 2 \end{cases} \xrightarrow{\text{Using online solver}} x_2 = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix} \quad \lambda_2 = 0.8$$

2) There aren't blocking constraints.

3) $A_2 = A_1$

KKT are satisfied. k1) All constraints are satisfied

k2) $\lambda_i > 0$

k3) $\lambda_{c1} = 0.8 \cdot 0 = 0$

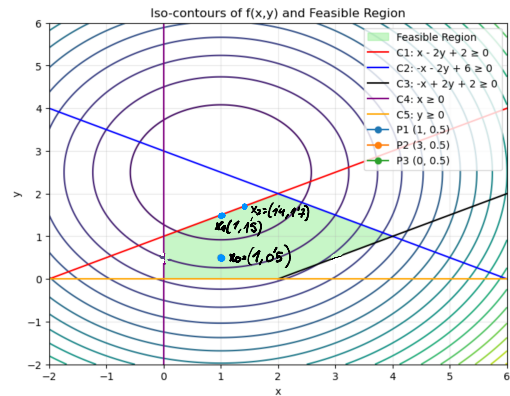
k4) $\nabla f(x^*) + 0.8 [-1 \ 2] = [2(1.4-1), 2(1.7-2.5)] + 0.8 [-1 \ 2] = 0$

ITER 3

$x_3 = x_2$ so the algorithm stops.

$$x^* = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(1, 1/2)	$\emptyset$	(1, 1.5)	-	$\{1\}$	NO
1	(1, 1.5)	$\{1\}$	(1.4, 1.7)	0.8	$\{1\}$	NO for x_1
2	(1.4, 1.7)	$\{1\}$	(1.4, 1.7)	0.8	END	YES
3						



2. $(x^0, y^0) = (3, 1/2)^T$ for $A^0 = \{3\}$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

$$\begin{cases} Qx_1 + \lambda \begin{bmatrix} 1 \\ -2 \end{bmatrix} = -\begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ \begin{bmatrix} 1 & -2 \end{bmatrix} x_1 = 2 \end{cases} \Rightarrow \begin{cases} 2x_1 + \lambda = 2 \\ 2x_1 - 2\lambda = 5 \end{cases} \xrightarrow{\text{using online solver}} x_1 = \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \lambda = -2.4$$

2) There aren't blocking constraints.

3) $A_1 = A_0$

KKT are NOT satisfied. k1) All constraints are satisfied ✓
k2) $\lambda_1 > 0$ XX

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

Same as previous: $x_2 = x_1$, $\lambda_2 = \lambda_1$.

Therefore $A_2 = \emptyset$

ITER 2

1) Solve KKT system $\begin{bmatrix} Q & B_2^T \\ B_2 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_2 \end{bmatrix}$

From previous case we know that $x_3 = \begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix}$ $\lambda_3 = 0.8$

2) The most blocking constraint is C_1 .

$$\alpha_j = \frac{b_j - \alpha_j^T x_k}{\alpha_j^T (x_{k+1} - x_k)} \quad \alpha_1 = \frac{2 - \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix}}{\begin{bmatrix} 1 & 2 \end{bmatrix} \left(\begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \right)} = \frac{2 + 2}{\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -0.8 \\ 2.4 \end{bmatrix}} = \frac{4}{6} = \frac{2}{3}$$

3) Update A_3 and x_3 :

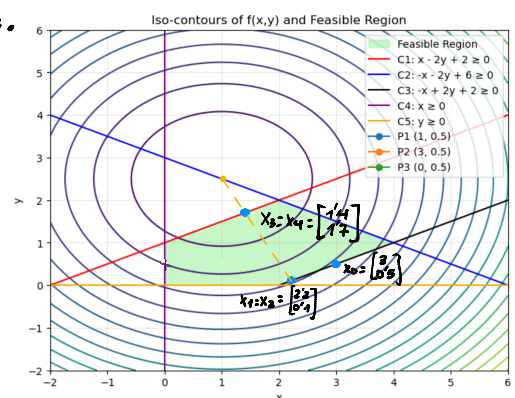
$$x_3 = x_2 + \alpha_1 (x_3 - x_2) = \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} + \frac{2}{3} \left(\begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \right) = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$$

$$A_3 = A_2 \cup \{1\} = \{1\}$$

From previous case we know that KKT are satisfied for x_3 .

ITER 3 → Stops because $x_4 = x_3$ and $\lambda_4 > 0$.

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(3, 0.5)	$\{3\}$	(2.2, 0.1)	-2.4	$\{3\}$	NO
1	(2.2, 0.1)	$\{3\}$	(2.2, 0.1)	-2.4	$\emptyset$	NO
2	(2.2, 0.1)	$\emptyset$	(1.4, 1.7)	0.8	$\{1\}$	NO for x_2
3	(1.4, 1.7)	$\{1\}$	(1.4, 1.7)	0.8	$\{1\}$	YES



3. $(x^0, y^0) = (0, 1/2)^T$ for $\mathcal{A}^0 = \{4\}$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda_1 \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

$$\begin{cases} Qx_1 + \lambda_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \\ [-1 \ 0] x_1 = 0 \end{cases} \Rightarrow \begin{cases} 2x_1 - \lambda_1 = 2 \Rightarrow \lambda_1 = -2 \\ 2x_1^2 = 5 \Rightarrow x_1^2 = 2.5 \\ -x_1 = 0 \Rightarrow x_1 = 0 \end{cases} \quad x_1 = \begin{bmatrix} 0 \\ 2.5 \end{bmatrix}$$

2) The most blocking constraint is C_1 .

$$\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)} \quad \alpha_1 = \frac{2 - [-1 \ 2] \begin{bmatrix} 0 \\ 2.5 \end{bmatrix}}{[-1 \ 2] (\begin{bmatrix} 0 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0.5 \end{bmatrix})} = \frac{2-1}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{1}{4}$$

3) Update A_1 and x_1 .

$$x_1 = x_0 + \alpha_1 (x_1 - x_0) = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} + \frac{1}{4} (\begin{bmatrix} 0 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A_1 = A_0 \cup \{1\} = \{4, 1\}$$

4) KKT not satisfied because $\lambda_1 < 0$

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

$$\begin{cases} Qx_2 + \lambda_2^1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \lambda_2^4 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ [-1 \ 2] x_2 = 2 \\ [-1 \ 0] x_2 = 0 \end{cases} \Rightarrow \begin{cases} 2x_2^1 - \lambda_2^1 - \lambda_2^4 = 2 \Rightarrow \lambda_2^4 = -1.5 - 2 = -3.5 \\ 2x_2^2 + 2\lambda_2^1 = 5 \Rightarrow 2 + 2\lambda_2^1 = 5 \Rightarrow \lambda_2^1 = 1.5 \\ -x_2^1 + 2x_2^2 = 2 \Rightarrow 2x_2^2 = 2 \Rightarrow x_2^2 = 1 \\ -x_2^1 = 0 \Rightarrow x_2^1 = 0 \end{cases}$$

2) As $x_2 = x_1$ and $\lambda_2^4 < 0$.

$$A_2 = A_1 - \{4\} = \{1\}$$

ITER 2

1) Solve KKT system $\begin{bmatrix} Q & B_2^T \\ B_2 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} -q \\ b_2 \end{bmatrix}$

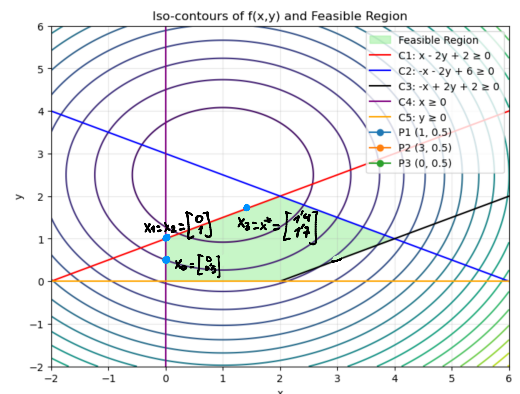
From previous case we know that $x_3 = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$ $\lambda_3 = 0.8$

2) There are not blocking constraints. x_3 is in C_1 .

3) As in the previous cases, KKT conditions are satisfied

ITER 3 $\rightarrow$ Stops b use $x_4 = x_3$ and $\lambda_4 > 0$.

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	$(0, 0.5)$	$\{4\}$	$(0, 1)$	-2	$\{1, 4\}$	NO
1	$(0, 1)$	$\{1, 4\}$	$(0, 1)$	$\lambda_2^1 = 1.5$ $\lambda_2^4 = -3.5$	$\{1\}$	NO
2	$(0, 1)$	$\{1\}$	$(1.4, 1.7)$	$\lambda_3^1 = 0.8$	$\{1\}$	YES
3	$(1.4, 1.7)$	$\{1\}$	$(1.4, 1.7)$	$\lambda_4 = 0.8$	END	YES



“Phase I” method for initial feasible point (0.25 pt)

We consider the original constraints:

$$\begin{aligned}(\text{C1}) \quad & x - 2y + 2 \geq 0, \\(\text{C2}) \quad & -x - 2y + 6 \geq 0, \\(\text{C3}) \quad & -x + 2y + 2 \geq 0, \\(\text{C4}) \quad & x \geq 0, \\(\text{C5}) \quad & y \geq 0.\end{aligned}$$

Given the starting point

$$x_0 = \begin{pmatrix} 1 \\ 2.5 \end{pmatrix},$$

we first check feasibility:

$$\begin{aligned}\text{C1: } & 1 - 2 \cdot 2.5 + 2 = -2 \quad (\text{violated by } 2), \\ \text{C2: } & -1 - 2 \cdot 2.5 + 6 = 0 \quad (\text{active}), \\ \text{C3: } & -1 + 2 \cdot 2.5 + 2 = 6 \quad (> 0), \\ \text{C4: } & 1 \geq 0, \quad \text{C5: } 2.5 \geq 0.\end{aligned}$$

Only Constraint (C1) is violated.

Phase I reformulation

Introduce the slack variable $s \geq 0$ to account for infeasibility and modify the constraints to follow the convention used in the lecture:

$$\begin{aligned}\min_{x,y,s} \quad & s \\ \text{s.t.} \quad & -x + 2y - 2 \leq s, \\ & x + 2y - 6 \leq 0, \\ & x - 2y - 2 \leq 0, \\ & -x \leq 0, \quad -y \leq 0, \quad s \geq 0.\end{aligned}$$

Choose $s_0 = 2 = \max(0, g_1(x_0))$ so that Constraint (C1) becomes satisfied at the starting point:

$$-1 + 2 \cdot 2.5 - 2 = 2.$$

Thus $(x_0, y_0, s_0) = (1, 2.5, 2)$ is feasible for the Phase I problem.

Solving Phase I

Because the original problem is feasible, the minimum of Phase I is $s^* = 0$, attained at any point satisfying (C1)–(C5).

A suitable feasible point is found by intersecting (C1) and (C2):

$$\begin{cases} -x + 2y - 2 = 0, \\ x + 2y - 6 = 0. \end{cases}$$

Solve the system:

$$\begin{aligned}x &= 2y - 2, \\ (2y - 2) + 2y - 6 &= 0 \Rightarrow 4y - 8 = 0 \Rightarrow y = 2,\end{aligned}$$

$$x = 2 \cdot 2 - 2 = 2.$$

Thus the candidate point is:

$$(x, y, s) = (2, 2, 0).$$

Check feasibility:

$$\text{C1: } 2 - 2 \cdot 2 + 2 = 0 \geq 0,$$

$$\text{C2: } -2 - 2 \cdot 2 + 6 = 0 \geq 0,$$

$$\text{C3: } -2 + 2 \cdot 2 + 2 = 4 \geq 0,$$

$$\text{C4: } 2 \geq 0, \quad \text{C5: } 2 \geq 0,$$

$$s = 0.$$

Thus, a valid feasible point for the original problem, produced by the Phase I method, is

$$x_{\text{feasible}} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

which satisfies all original constraints and can be used to start the active-set method.

”Big M” method for initial feasible point (Bonus 0.5 pt)

We introduce the variable

$$z := \begin{pmatrix} x \\ y \\ s \end{pmatrix} \in \mathbb{R}^3$$

and rewrite the Big M formulation as a quadratic program of the form

$$\min_z \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{subject to} \quad Bz \leq b.$$

The Big M objective is

$$F_M(x, y, s) = (x - 1)^2 + (y - 2.5)^2 + 100s,$$

so

$$Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad q = \begin{pmatrix} -2 \\ -5 \\ 100 \end{pmatrix}, \quad r = 7.25.$$

The constraints

$$-x + 2y - 2 \leq s,$$

$$x + 2y - 6 \leq 0,$$

$$x - 2y - 2 \leq 0,$$

$$-x \leq 0, \quad -y \leq 0, \quad s \geq 0$$

are written as

$$Bz \leq b, \quad B \in \mathbb{R}^{6 \times 3}, \quad b \in \mathbb{R}^6,$$

with rows

$$B = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 1 & -2 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 6 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The starting point is

$$z_0 = \begin{pmatrix} 1 \\ 2.5 \\ 2 \end{pmatrix},$$

for which

$$Bz_0 - b = \begin{pmatrix} 0 \\ 0 \\ -6 \\ -1 \\ -2.5 \\ -2 \end{pmatrix} \leq 0,$$

so z_0 is feasible and constraints 1 and 2 are active:

$$\mathcal{A}_0 = \{1, 2\}.$$

Active-set method for the Big M QP

For a given active set $\mathcal{A}_k$, let B_k and b_k collect the corresponding rows of B and entries of b . The active-set subproblem is

$$\min_z \frac{1}{2} z^\top Q z + q^\top z + r \quad \text{s.t.} \quad B_k z = b_k,$$

whose KKT system is

$$\begin{pmatrix} Q & B_k^\top \\ B_k & 0 \end{pmatrix} \begin{pmatrix} z_{k+1}^{\text{cand}} \\ \lambda_{k+1, \mathcal{A}_k} \end{pmatrix} = \begin{pmatrix} -q \\ b_k \end{pmatrix}.$$

This gives a candidate point z_{k+1}^{cand} ; the search direction is $p_k = z_{k+1}^{\text{cand}} - z_k$, and we choose a step length $\alpha_k \in (0, 1]$ so that $z_k + \alpha_k p_k$ remains feasible. If $\alpha_k < 1$, a new constraint becomes active. If $p_k = 0$, we inspect the multipliers and possibly drop a constraint with negative multiplier.

Iteration 0. Active set: $\mathcal{A}_0 = \{1, 2\}$, so

$$B_0 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \end{pmatrix}, \quad b_0 = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_0^\top \\ B_0 & 0 \end{pmatrix} \begin{pmatrix} z_1^{\text{cand}} \\ \lambda_{1, \mathcal{A}_0} \end{pmatrix} = \begin{pmatrix} -q \\ b_0 \end{pmatrix}$$

yields

$$z_1^{\text{cand}} = \begin{pmatrix} 81 \\ -37.5 \\ -158 \end{pmatrix}, \quad \lambda_{1,1} = 100, \quad \lambda_{1,2} = -60.$$

The search direction is

$$p_0 = z_1^{\text{cand}} - z_0 = \begin{pmatrix} 80 \\ -40 \\ -160 \end{pmatrix}.$$

The inactive constraints along $z(\alpha) = z_0 + \alpha p_0$ satisfy

$$\begin{pmatrix} 160\alpha - 6 \\ -80\alpha - 1 \\ 40\alpha - 2.5 \\ 160\alpha - 2 \end{pmatrix} \leq 0,$$

which gives

$$\alpha_0 \leq \min \left\{ \frac{6}{160}, \frac{2.5}{40}, \frac{2}{160} \right\} = \frac{1}{80}.$$

The blocking constraint is 6, so $\alpha_0 = 1/80$ and

$$z_1 = z_0 + \alpha_0 p_0 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

At z_1 , $Bz_1 - b = (0, 0, -4, -2, -2, 0)^\top$, so

$$\mathcal{A}_1 = \{1, 2, 6\}.$$

Iteration 1. With $\mathcal{A}_1 = \{1, 2, 6\}$,

$$B_1 = \begin{pmatrix} -1 & 2 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}.$$

Solving

$$\begin{pmatrix} Q & B_1^\top \\ B_1 & 0 \end{pmatrix} \begin{pmatrix} z_2^{\text{cand}} \\ \lambda_{2,\mathcal{A}_1} \end{pmatrix} = \begin{pmatrix} -q \\ b_1 \end{pmatrix}$$

gives

$$z_2^{\text{cand}} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = z_1, \quad \lambda_{2,1} = \frac{5}{4}, \quad \lambda_{2,2} = -\frac{3}{4}, \quad \lambda_{2,6} = \frac{395}{4}.$$

Hence $p_1 = 0$. Since one multiplier is negative ($\lambda_{2,2} < 0$), constraint 2 is removed from the active set:

$$\mathcal{A}_2 = \{1, 6\}, \quad z_2 := z_1.$$

Iteration 2. Now

$$B_2 = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

The KKT system

$$\begin{pmatrix} Q & B_2^\top \\ B_2 & 0 \end{pmatrix} \begin{pmatrix} z_3^{\text{cand}} \\ \lambda_{3,\mathcal{A}_2} \end{pmatrix} = \begin{pmatrix} -q \\ b_2 \end{pmatrix}$$

yields

$$z_3^{\text{cand}} = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}, \quad \lambda_{3,1} = \frac{4}{5}, \quad \lambda_{3,6} = \frac{496}{5}.$$

The search direction is

$$p_2 = z_3^{\text{cand}} - z_2 = \begin{pmatrix} -\frac{3}{5} \\ -\frac{3}{10} \\ 0 \end{pmatrix}.$$

Along $z(\alpha) = z_2 + \alpha p_2$, the inactive constraints satisfy

$$\begin{pmatrix} -\frac{6}{5}\alpha \\ -4 \\ \frac{3}{5}\alpha - 2 \\ \frac{3}{10}\alpha - 2 \end{pmatrix} \leq 0$$

for all $\alpha \in [0, 1]$. Thus no new constraint blocks the step and we take $\alpha_2 = 1$:

$$z_3 = z_2 + p_2 = \begin{pmatrix} \frac{7}{5} \\ \frac{17}{10} \\ 0 \end{pmatrix}.$$

At z_3 , $Bz_3 - b = (0, -\frac{6}{5}, -4, -\frac{7}{5}, -\frac{17}{10}, 0)^\top$, so

$$\mathcal{A}_3 = \{1, 6\}.$$

Iteration 3 (optimality). Solving the KKT system again with $\mathcal{A}_3 = \{1, 6\}$ returns

$$z_4^{\text{cand}} = z_3, \quad \lambda_{4,1} = \frac{4}{5}, \quad \lambda_{4,6} = \frac{496}{5},$$

so $p_3 = 0$ and all multipliers of active constraints are nonnegative. Therefore the KKT conditions are satisfied and the active-set method terminates at

$$(x^*, y^*, s^*) = \left(\frac{7}{5}, \frac{17}{10}, 0 \right) = (1.4, 1.7, 0).$$

This matches the solution we found in the first exercise as expected.